Choice Reaction Time and Response Inhibition - Insight46 phase 1

Kirsty Lu | 29/07/2019

Document details

Categories of variables	Choice Reaction Time and Response Inhibition
Summary of work undertaken	1. Data collected, processed and cleaned as described below. 2. Summary outcomes calculated as described below.
Names of source variables	For, descriptions, of, what, these, variables, mean, see, the, labels, in, the, dta, file, id, ID, number), Scomp, Swn
Output variables	circle_comp_i46p1, circle_wn_i46p1, circle_rateo_i46p1, circle_dual_i46p1, circle_single_comp_i46p1, circle_numrotd_i46p1, circle_numdev_rotd_i46p1, circle_toto_i46p1, circle_errn_i46p1, circle_erro_i46p1, circle_dualwn_i46p1, reaction_numtrials_i46p1, reaction_err_rate_i46p1, reaction_err_num_i46p1, reaction_mrt_i46p1, reaction_rtsd_i46p1, reaction_var_i46p1, resp_inhib_wn_i46p1, resp_inhib_numtrials_i46p1, resp_inhib_err_rate_i46p1, resp_inhib_mrt_i46p1, resp_inhib_rtcong_i46p1, resp_inhib_rtincong_i46p1, resp_inhib_err_cong_i46p1, resp_inhib_err_inc_i46p1
Papers using these variables	Characterising cognition in later life and its relationship with biomarkers of Alzheimer's disease: a study of members of the National Survey of Health and Development (the British 1946 birth cohort). PhD thesis (Kirsty Lu) **It is recommended that potential users of these datasets discuss their analyses with Kirsty Lu.**

Overview

<u>For Submission to the NSHD Scientific Support Team</u>

Background

A brief description has been published in the Insight 46 protocol paper:

Lane et al. (2017) Study protocol: Insight 46 – a neuroscience sub-study of the MRC National Survey of Health and Development, BMC Neurology, 17:75, doi: 10.1186/s12883-017-0846-x

Two related tasks – Choice Reaction Time (RT) and Response Inhibition – were combined into one experiment, presented on a DELL Optiplex 9030 all-in-one touchscreen computer, using SuperLab software. A response box with two buttons side by side was connected to the computer and placed on the table in front of the participant.

The **Choice RT** task had 2 blocks of 12 trials each. In block 1 the stimulus was an arrow pointing left or right, and in block 2 the stimulus was a word – 'LEFT' or 'RIGHT' (see Figure 6-1). All stimuli were displayed in the centre of the screen. At the beginning of each trial, there was an interval of 1000ms throughout which a cue was displayed ('Arrow' in Block 1, and 'Word' in Block 2), before the stimulus appeared below it. The purpose of displaying the cue before the stimulus was to pace the experiment and to maintain a consistent format with the second experiment (see Chapter 7) where cues formed a necessary part of the instructions.

The **Response Inhibition** task were 2 blocks of 48 trials each. In Block 1, the cue duration was 200ms (i.e. the cue was displayed on its own for 200ms before the stimulus appeared below it). In Block 2 the cue duration was 1500ms. Throughout both blocks, the order of the trials was A-A-A-W-W-W-A-A-A-W-W-W etc., where A indicates an arrow cue and W indicates a word cue. Throughout both blocks, the order of the trials was A-A-A-W-W-W-A-A-A-W-W-W etc., where A indicates an arrow cue and W indicates a word cue. Thus, there were equal numbers of arrow cue and word cue trials. Within each block there were equal numbers of trials in each of the four possible combinations of congruency and cue type (congruent arrow, incongruent arrow, congruent word, incongruent word). Six practice trials preceded each block. The practice trials were followed by a gap of about 5 seconds before the main block started automatically.

For **both tasks**, participants were instructed to press the correct button on the response box (left or right) as quickly as possible, according to the stimulus on the screen. They were asked to use the index and middle fingers of their dominant hand, with one finger on each button. If their response was correct, the next trial was initiated; if their response was incorrect, the stimulus remained on the screen and an error tone sounded to signal that they should respond again as quickly as possible. Regardless of whether their second attempt was correct or incorrect, no feedback was given and the next trial was initiated. Six practice trials preceded each block. The practice trials were followed by a gap of about 5 seconds before the main block started automatically.

Data processing and cleaning

Each participant's spreadsheet was imported into Stata and all were merged into one dataset. The dataset was cleaned to remove all practice trials and all rows not pertaining to the first response to each trial. The dataset was then split into two, separating the Choice Reaction Time task (blocks 1 & 2) and Response Inhibition task (blocks 3 & 4) into two separate datasets. For the **Choice RT** dataset, the following cleaning and coding was carried out: 1. Trials were numbered in the order administered (Rtrial) 2. Trials were coded as arrow or word (Raw) 3. The variable Rcorr was created to code what the correct response to each trial would be (left or right). 4. The variable Rerr was created to code whether each response was correct or incorrect 5. Some visit numbers were incorrectly entered as "1946" rather than 1, so this was amended. 6. Reaction time (Rrt) was calculated by subtracting 1000ms from the times imported by the spreadsheet, since this time included the 1000ms period during which the cue was displayed, before the stimulus appeared. 7. The first trial of each block was dropped as participants were not always ready for it, due to limitations of the experimental design. 8. A threshold of 300ms was applied as the fastest time for a genuine response - in fact there were no responses below this threshold, so no responses were dropped on this basis. 9. In order to exclude outlying slow responses, each participant's mean RT and SD were calculated, based on all responses (correct and incorrect). Trials were dropped if the response was greater than 3 SD above the individual's mean. This applied to 40 responses from 40 participants (1 response each). 10. One participant was dropped as they were noted to have difficulty in understanding the task and the tester did not believe their data were valid.

This dataset was saved as Insight46_choiceRT_fulldata_cleaned_final_20190729 Note that it contains only the 501 participants with useable data from the choice RT task. For details of the participant who did not have useable data – and reasons why – see the summary dataset described below.

A second dataset was created which contains one row per participant, summarising their performance across the Choice RT task. The following variables were calculated for each participant: 1. A variable was created to indicate how many useable trials each participant had (Rnumtrials): in most cases this was 22, but it was 21 for the participants with a trial excluded (see above). 2. A variable was created to indicate which participants had usable data from the Choice RT task (Rusable). (One participant did not, as explained above) 3. Mean RT for correct responses (Rmrt) 4. Error rate, i.e. proportion of incorrect responses (Rerr_rate) 5. Number of errors (Rerr_num) 6. Standard deviation of RT for correct responses (Rrtsd) 7. Intra-individual variability in RT (Rvar), which is the coefficient of variation i.e. SD/mean (Rrtsd/Rmrt)

This dataset was saved as Insight46_choiceRT_summaryoutcomes_cleaned_final_20190729 Note that it contains all 502 Insight46 participants.

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This dataset was saved as Insight46_choiceRT_fulldata_cleaned_final_20190729 Note that it contains only the 500 participants with useable data from the response inhibition task.

A second dataset was created which contains one row per participant, summarising their performance across the Choice RT task. The following variables were calculated for each participant: 1. A variable was created to indicate how many useable trials each participant had (Snumtrials): the maximum was 94 but it was lower for the participants with one or more trials excluded (see above). 2. A variable was created to indicate which participants had usable data from the Response Inhibition task (Susable). (Two participants did not, as explained above) 3. Mean RT for correct responses (Smrt) 4. Error rate, i.e. proportion of incorrect responses (Serr_rate) 5. Mean RT for correct responses to congruent trials (Srtcong) 6. Mean RT for correct responses to incongruent trials (Srtincong) 7. Error rate on congruent trials (Serr_ratecong) 8. Error rate on incongruent trials (Serr_rateincong)

This dataset was saved as Insight46_responseinhibition_summaryoutcomes_cleaned_final_20190830 Note that it contains all 502 Insight46 participants.

Variables in this document

22 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Reaction time [60423]		
reaction_err_num_i46p1	Choice reaction time (RT): number of errors - Insight46 phase 1	topsheet field
reaction_err_rate_i46p1	Choice reaction time (RT): Error rate (%) - Insight46 phase 1	topsheet field
reaction_mrt_i46p1	Choice reaction time (RT): Mean RT for correct trials - Insight46 phase 1	topsheet field
reaction_numtrials_i46p1	Choice reaction time (RT): number of trials with usable RT data - Insight46 phase 1	topsheet field
reaction_rtsd_i46p1	Choice reaction time (RT): Standard devision of reaction time for correct trials - Insight46 phase 1	topsheet field
reaction_var_i46p1	Choice reaction time (RT): Intra-individual variability in reaction time (RT) for correct responses (Standard Deviation/mean) - Insight46 phase 1	topsheet field
resp_inhib_err_cong_i46p1	Switching: Error rate for congruent trials - Insight46 phase 1	topsheet field
resp_inhib_err_inc_i46p1	Switching: Error rate for incongruent trials - Insight46 phase 1	topsheet field
resp_inhib_err_rate_i46p1	Switching: Each participant's overall error rate - Insight46 phase 1	topsheet field
resp_inhib_mrt_i46p1	Switching: Mean reaction time (RT) (correct responses only) - Insight46 phase 1	topsheet field
resp_inhib_numtrials_i46p1	Switching: number of trials with usable response time (RT) data - Insight46 phase 1	topsheet field
resp_inhib_rtcong_i46p1	Switching: Mean reaction time (RT) for congruent trials (correct responses only) - Insight46 phase 1	topsheet field

Variable	Label	How it was found
resp_inhib_rtincong_i46p1	Switching: Mean reaction time (RT) for incongruent trials (correct responses only) - Insight46 phase 1	topsheet field
Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Visuomotor performance [60427]		
circle_comp_i46p1	Circle tracing: was the task completed? - Insight46 phase 1	topsheet field
circle_dual_i46p1	Circle tracing: participant has usable dual-task data - Insight46 phase 1	topsheet field
circle_errn_i46p1	Circle tracing: total number of incorrect subtraction responses summed across all 6 trials - Insight46 phase 1	topsheet field
circle_erro_i46p1	Circle tracing: percentage of incorrect subtraction responses summed across all 6 trials combined - Insight46 phase 1	topsheet field
circle_numdev_rotd_i46p1	Circle tracing: mean number of errors per rotation across all dual-task trials - Insight46 phase 1	topsheet field
circle_numrotd_i46p1	Circle tracing: mean number of rotations across all dual-task trials - Insight46 phase 1	topsheet field
circle_rateo_i46p1	Circle tracing: mean subtraction rate (responses per second) - Insight46 phase 1	topsheet field
circle_single_comp_i46p1	Circle tracing: Participant completed the single-task circle-tracing trials - Insight46 phase 1	topsheet field
circle_toto_i46p1	Circle tracing: total number of subtraction responses summed across all 6 trials - Insight46 phase 1	topsheet field

2 low-confidence matches

Variable	Label	How it was found
Clinical data [51] › Physical measures [110] › Hand examination [1106]		
hand	Now I would like to examine your hands for any bumps or swellings. Would you be willing for me to examine your hands? - first record - at age 53 years	prose scan
Questionnaire data [55] › Sociodemographics [515] › Household [5152]		
own	Now I would like to get some general information about your household. Does your household own or rent this accommodation? - at age 53 years	prose scan